

AI-Powered Document Intelligence System using AWS RAG

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Abstract

The rapid growth of digital documents in organizations and educational institutions has created significant challenges in efficiently retrieving and understanding information from unstructured data sources. Traditional document search systems primarily rely on keyword-based retrieval techniques, which often fail to provide accurate, context-aware, and intelligent responses. This limitation reduces the efficiency of knowledge extraction and document analysis in real-world applications.

This project introduces a cloud-based AI-powered document retrieval and question-answering system that utilizes Retrieval-Augmented Generation (RAG) architecture to enable intelligent interaction with uploaded documents. The system allows users to upload PDF documents through an interactive web interface and ask natural language questions related to the uploaded content. The uploaded files are securely stored in Amazon S3 and processed using Amazon Textract to extract textual information from documents in real time. The extracted content is transformed into semantic vector embeddings using Amazon Bedrock Titan Embeddings and indexed in Amazon OpenSearch Service for efficient semantic search and contextual information retrieval. The application further integrates Amazon Bedrock Claude models to generate accurate and context-aware responses based on the retrieved document content. The complete workflow is implemented using a scalable serverless cloud architecture consisting of AWS Lambda, API Gateway, Amazon S3, Amazon Bedrock, and OpenSearch Service.

By leveraging cloud computing, generative AI, and semantic search technologies, the proposed system delivers a scalable, reliable, and intelligent platform for automated document analysis and knowledge retrieval. The solution demonstrates the practical integration of modern AI services and cloud-native infrastructure to build enterprise-level intelligent assistants for educational, research, and organizational environments. The project also highlights the effectiveness of serverless architectures in developing scalable and cost-efficient AI-driven applications.

Keywords

RAG, Document Intelligence, Semantic Search, AWS, Amazon Bedrock, OpenSearch, Textract, Serverless Architecture, NLP, Generative AI

Problem Statement

The exponential increase in digital documentation has made it difficult for users to efficiently retrieve relevant information. Existing systems suffer from several key limitations:

1. Inefficient Keyword-Based Search

Traditional systems rely on exact keyword matching, often missing the contextual meaning of queries.

2. Lack of Context Awareness

Search engines fail to understand user intent, leading to irrelevant or incomplete results.

3. Manual Document Analysis

Users must manually read large volumes of text, which is time-consuming and inefficient.

4. Poor Knowledge Extraction

Unstructured data remains underutilized due to lack of intelligent processing tools.

Proposed Solution

To address these challenges, the proposed system introduces an AI-driven document intelligence platform that:

- Accepts PDF document uploads via a user-friendly web interface
- Extracts text using Amazon Textract
- Converts extracted content into semantic embeddings using Bedrock Titan
- Stores and indexes embeddings in OpenSearch for fast retrieval
- Uses RAG architecture to combine retrieval with generation
- Generates accurate answers using Bedrock Claude models
- Provides context-aware responses to user queries

This solution ensures efficient, intelligent, and scalable document interaction.

SDG Alignment

SDG Goal	Target	Relevance to Project
SDG 4 — Quality Education	Target 4.4	Enhances access to knowledge by enabling intelligent document search and automated question answering for students and researchers.
SDG 9 — Industry, Innovation and Infrastructure	Target 9.5	Promotes innovation by integrating AI, cloud computing, and semantic search technologies for advanced information retrieval systems.
SDG 16 — Peace, Justice and Strong Institutions	Target 16.10	Improves access to information by providing accurate, reliable, and context-aware document insights.

SDG Goal	Target	Relevance to Project
SDG 8 — Decent Work and Economic Growth	Target 8.2	Increases productivity by reducing time spent on manual document analysis and improving workflow efficiency.

System Architecture & AWS Services

Service	Role
Amazon S3	Stores uploaded documents
Amazon Textract	Extracts text from PDFs
AWS Lambda	Handles backend processing
API Gateway	Manages API requests
Amazon Bedrock	Generates embeddings & responses
OpenSearch Service	Stores and retrieves vector data

Workflow

1. User uploads a document via web interface
2. File is stored in Amazon S3
3. Textract extracts text from document
4. Text is converted into embeddings
5. Embeddings are stored in OpenSearch
6. User submits a query
7. Relevant content is retrieved
8. Claude model generates final answer

Key Features

- Natural language question answering
- Semantic search instead of keyword search
- Real-time document processing

- Scalable serverless architecture
- Secure cloud storage
- Context-aware AI responses

Advantages

- Reduces manual effort in document analysis
- Improves accuracy of information retrieval
- Handles large-scale document datasets
- Cost-efficient due to serverless design
- Easy integration with enterprise systems

Applications

- Educational content analysis
- Research paper summarization
- Legal document review
- Business knowledge management
- Healthcare record analysis

Future Enhancements

- Support for multiple file formats (DOCX, images, etc.)
- Multi-language query support
- Voice-based interaction
- Fine-tuned domain-specific models
- Real-time collaboration features

Conclusion

The AI-Powered Document Intelligence System using AWS RAG provides an advanced solution for intelligent document retrieval and analysis. By combining semantic search with generative AI, the system overcomes the limitations of traditional keyword-based methods. Its scalable cloud-native architecture ensures reliability and performance, making it suitable for real-world enterprise applications. The project highlights the transformative potential of integrating AI and cloud technologies for efficient knowledge management.